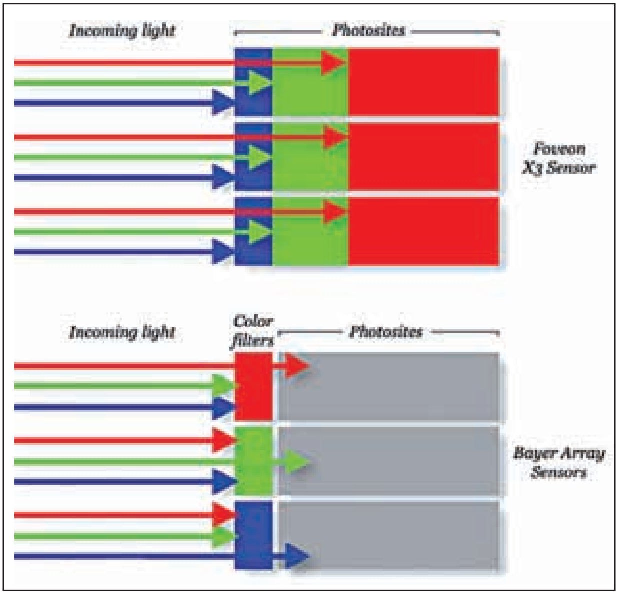

A visual representation of the difference in arrangement

filter or the Red-Green-Blue-Emerald (RGBE) filter. Another option is the Fovean X3 sensor, where demosaicing is not required. It's a CMOS sensor that instead of forming a mosaic out of them layers red, green and blue sensors vertically, forming an array of photo-sites, each of which consists of three vertically stacked photodiodes, organised in a two-dimensional grid. The signals from the photodiodes are then processed to extract data that provides the three additive primary colours: red, green and blue. Less popular than the Bayer filter image sensors for digital cameras, the Fovean X3s differ in operation. Every photo-site in Bayer sensors consists of a single light sensor, be it CMOS or CCD, which is exposed to only one of the three primary colours due to filtration. This raises the need of using an interpolative process like demosaicing to get the full-colour image, where each photo-site's output pixel is assigned an RGB value which depends on the levels of red, blue and green reported by the nearby photo-sites. On the other hand, a Fovean X3 combines the output from all the stacked photodiodes at each of its photo-sites to create a final RGB colour result for every site.

And it's this simple operational difference which leads to consequences making one option more preferred than the other for a particular kind of shoot. The absence of demosaicing in Fovean X3 has its own advantages.